Modelling

National 5 · Design · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ explain how models are used to generate and explore, test and refine, and communicate
- ☐ give reasons for choosing sketch, scale, block and computer-generated models
- ☐ describe the advantages of using models in the design process

I DO — Worked example

Read how a full-mark answer is built

A designer developed a hand-held torch.

1 Describe one benefit of a sketch model and one benefit of a block model for the torch. (4 marks)

Full-mark answer each ✓ is one mark

Sketch model: it can be made quickly from card or foam, so the designer can try several different torch shapes in a lesson ✓ and see forms that are hard to draw ✓. Block model: it shows the true size and shape of the torch so it can be held to test the grip and the position of the button ✓, and it can be shown to the client for feedback ✓.

Why answers lose marks

"Sketch models are cheap and quick. Block models are good to see what it looks like."

0 marks: unqualified "cheap/quick", and "see what it looks like" is too vague.

WE DO — Guided practice

Use the starters and word bank to help you

2 Explain two reasons a designer would use a 3D CAD model. (2 marks)

Sentence starters

One reason is … because …

Another reason is … so …

Word bank

change · starting · Prototype · without · 3D printer · client · viewed · rendered

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Why are sketch models used early in the design process?
☐ They show the final finish ☐ They are quick and cheap to make and change ☐ They are fully working ☐ They are used for mass production

b) Which model would a designer use to test the comfort of a handle grip?
☐ Scale model ☐ Block model ☐ Sketch model of a different product ☐ Exploded drawing

c) An architect shows a new school building to the council. Which model?
☐ Sketch model ☐ Scale model ☐ Test rig ☐ Prototype

4 Key terms: write the letter of the matching description next to each term.

1. **Sketch model**	____	**A** Solid, non-working model showing shape and form; often used to test ergonomics.
2. **Block model**	____	**B** Quick, cheap model from scrap materials to generate and explore ideas.
3. **Scale model**	____	**C** Model at a different scale for objects too large or small to model full size.

5 Describe one benefit of a computer-generated model. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?